FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

IT0049
(WEB SYSTEM TECHNOLOGIES)

TECHNICAL FORMATIVE ASSESSMENT

6

FROM ARRAYS TO ELOQUENT

Student Name / Group Name:	Rafael Miguel N. Joaquin
Members (if Group):	**Name** / **Role**
Section:	TX32

Professor:	Von Erick Magbitang

PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements. [PO: C]

COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- Apply advanced web development principles in creating, debugging, validating, and securing database-backed web applications. [CLO: 2]

INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- Create Laravel migrations for the customers and users tables.
- Create and run a seeder to populate sample data.
- Build Eloquent models for both tables.
- Retrieve and display records in a Blade view.

BACKGROUND INFORMATION

This activity replicates TFA2 in Laravel. Instead of hand-writing SQL to create tables, Laravel migrations describe the schema as PHP code that can be run, rolled back, and shared through version control. A seeder then populates sample data automatically, and Eloquent — Laravel's ORM — retrieves records as PHP objects without raw queries.

GRADING SYSTEM / RUBRIC

CRITERIA	Description
Functionality & Requirements Completeness (40 points max)	All required pages and features are present and work as specified in the activity instructions.
Code Structure & Organization (25 points max)	Controllers, models, and views are separated correctly and follow the framework's conventions; code is readable and sensibly named.
Database Design & Query Usage (20 points max)	Migrations correctly create the specified schema, the seeder populates realistic sample data, and records are retrieved through Eloquent models and displayed correctly in the view.

CRITERIA	Description
Documentation & Submission Quality (15 points max)	The GitHub repository is organized and complete, the README explains how to set up and run the project, and the hosted link is live and matches the submitted code.

LABORATORY ACTIVITY

Rebuild TFA5's POS application with a real database: replace the static arrays with migrations, a seeder, and Eloquent models, using the same schema as TFA2.

1. Create migrations for the customers table (full_name, email, phone) and the users table (username unique, full_name), matching TFA2's schema.
2. Create a seeder that populates at least 5 sample records in each table.
3. Create Customer and User Eloquent models.
4. Update your CustomerController and UserController to retrieve records through their Eloquent models instead of static arrays.
5. Confirm the Blade views display the retrieved records the same way they displayed the static array data in TFA5.

Submission Requirements:

- A link to a GitHub repository containing your raw project files, including your database export or migration and seeder files.
- A link to your hosted, working version of the application.

REFERENCES

- Stauffer, M. (2024). Laravel: Up & Running: A Framework for Building Modern PHP Apps (3rd ed.).
- Laravel Documentation. Migrations. https://laravel.com/docs/13.x/migrations
- Laravel Documentation. Eloquent — Getting Started. https://laravel.com/docs/13.x/eloquent
- Laravel Documentation. Seeding. https://laravel.com/docs/13.x/seeding